

LandSetu: An AI-Driven National Land Governance Intelligence Platform Integrating Heterogeneous Land Records and Public Grievances for Evidence-Based Policy Decision-Making

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[Live Prototype: sih-proto-frontend.vercel.app](https://sih-proto-frontend.vercel.app)

Abstract—Every year, millions of Indian families wait for a single stamped signature to tell them a piece of land is truly theirs — yet land-related litigation is estimated to occupy a majority share of pending civil cases in Indian courts, even as over 95% of rural land records now sit digitized in government servers. The paradox is structural: land revenue departments, survey and settlement records, geospatial cadastral datasets, and citizen grievance platforms operate as isolated silos, leaving administrators unable to see, in one place, which regions suffer recurring disputes, which mutation requests are stuck the longest, or which policy gaps are driving citizen dissatisfaction. This paper presents LandSetu, an AI-driven national land governance intelligence platform that harmonizes structured land records with unstructured grievance narratives through entity resolution, geospatial alignment, OCR-based document digitization, and natural language processing. The platform's processing core combines data integration, machine learning classification, spatial analysis, retrieval-augmented legal knowledge lookup (RAG), and policy-impact simulation to convert fragmented records into an evidence-based policy dashboard, supported by a citizen-facing grievance portal and a real-time officer dashboard synchronized with government databases. A working prototype was implemented and evaluated on a small pilot sample of representative grievance records and scanned land documents. On this pilot sample ($n = 60$), the LandSetu NLP classification pipeline achieved 83.3% accuracy, 81.0% precision, 80.0% recall, and an 80.5% F1-score, compared with 68.3% accuracy for a keyword/rule-based baseline. The results, while based on a small-scale pilot rather than a production deployment, indicate that integrating previously siloed land-governance data sources through AI can meaningfully improve the detection of systemic bottlenecks and support transparent, targeted policy intervention.

Keywords—*Land Governance, Data Integration, Natural Language Processing, Entity Resolution, Public Grievances, Evidence-Based Policy, Retrieval-Augmented Generation, Geospatial Analysis, e-Governance.*

I. INTRODUCTION

Somewhere in rural India today, a farmer is standing in line outside a tehsil office for the third time this month, holding a mutation application that has quietly gone nowhere. Multiply that single wait by the estimated tens of millions of pending land-related cases clogging Indian courts, and land governance stops looking like a paperwork problem and starts looking like a national productivity crisis — one that touches credit access, agricultural investment, urban planning, and social trust in the state itself.

India has not been passive about this. The Digital India Land Records Modernization Programme (DILRMP), launched in 2016, has computerized over 95% of rural land records and digitized a majority of cadastral maps nationwide [6]. The government has gone further still, assigning every land parcel a unique 14-digit Unique Land Parcel Identification Number (ULPIN) — popularly nicknamed “Bhu-Aadhaar” — so that, in principle, no parcel of Indian soil need ever again be ambiguous about who owns it [7].

And yet the disputes persist. The reason is not a lack of data but a lack of connection between data. Revenue records live in one department's database, survey and settlement maps in another, geospatial cadastral layers in a third, and citizen grievances — arguably the richest, most human signal of where the system is failing — sit almost entirely unread in free-text form on grievance portals. No administrator today can easily ask, and get an evidence-based answer to, a question as simple as: “Which district has both slow mutation processing and a rising cluster of title-dispute complaints, and why?”

This paper introduces LandSetu, a national land governance and research intelligence platform designed to close exactly that gap. Rather than digitizing more data, LandSetu is built to make sense of the data that already exists — fusing structured revenue and survey records, OCR-scanned legacy documents, geospatial layers, and unstructured citizen grievances into a single AI-driven analytical layer that surfaces patterns no single department could see on its own. The remainder of this paper defines the problem precisely, positions LandSetu against the existing research and product landscape, details its architecture and methodology, and reports preliminary results from an early pilot evaluation of the prototype.

II. PROBLEM STATEMENT

Existing land governance systems in India digitize records within isolated departmental silos and therefore lack a mechanism to correlate structured revenue/survey data with unstructured citizen grievance narratives at the parcel or regional level. Consequently, decision-makers cannot systematically identify which specific governance processes

are generating clusters of disputes, which regions show abnormally long mutation-processing delays, or which grievance themes require the most urgent policy attention, resulting in reactive rather than evidence-based land administration.

III. RESEARCH GAP

A review of the existing landscape reveals four recurring gaps that this work aims to address:

- Prior GIS-based land record systems focus primarily on spatial visualization and title-mapping, without integrating citizen grievance text as an analytical input.
- Existing grievance-analytics platforms classify and route complaints but rarely link them back to structured land-parcel or survey records for root-cause analysis.
- Few frameworks apply entity resolution to reconcile citizen-reported locations/parcel references with authoritative revenue and geospatial datasets, or use retrieval-augmented legal knowledge lookup to ground insights in actual law and precedent.
- Most current systems produce descriptive dashboards (counts, status trackers) rather than correlative, simulation-backed, policy-actionable insights connecting governance-process gaps to dispute patterns.

IV. OBJECTIVES

The specific objectives of this research are:

- To design a data harmonization framework that unifies structured land revenue/survey records with unstructured citizen grievance text and scanned legacy documents using entity resolution, OCR, and geospatial alignment.
- To develop an NLP-based pipeline for grievance classification and theme extraction from citizen complaint narratives.
- To integrate a retrieval-augmented legal knowledge layer capable of grounding insights in applicable laws, policies, and court rulings.
- To build a correlation and policy-impact simulation module linking governance-process indicators with dispute and delay patterns.
- To implement and evaluate a working prototype (LandSetu) on a representative pilot dataset using standard performance metrics.

V. RESEARCH QUESTIONS

RQ1: How can heterogeneous land-governance data sources — including scanned legacy documents — be harmonized into a unified, parcel-level schema using entity resolution and geospatial alignment?

RQ2: How effectively can transformer-based NLP techniques classify and extract themes from unstructured citizen grievance narratives compared with a rule-based baseline?

RQ3: What correlations exist between specific land-governance process gaps and the frequency or persistence of disputes and delays in a given region?

RQ4: Can a retrieval-augmented legal knowledge layer and policy-impact simulation module make the resulting insights directly actionable for administrators, rather than merely descriptive?

VI. LITERATURE REVIEW AND COMPARATIVE ANALYSIS

Several recent studies have explored AI-assisted citizen grievance handling. Aakash et al. proposed CivicSense, a generative-AI and NLP-based platform that automatically identifies, ranks, and summarizes large volumes of municipal citizen complaints, converting unstructured text into structured, actionable insight for authorities [1]. Kotwal et al. developed a web-based Grievance Redressal Portal and explicitly noted that existing systems suffer from an absence

of unified platforms and minimal analytical insight for administrators, a gap this paper directly targets [2]. A study conducted for Nairobi County used a support vector classifier with TF-IDF features to categorize social-media civic complaints, achieving 77.52% classification accuracy, and applied named-entity recognition with reverse geocoding to spatially locate complaints [3]. An ACL workshop study on complaint analysis for economic and food-safety regulators applied traditional feature-based classifiers to citizen feedback, reporting accuracy scores above 70% [4]. A Flask/MySQL-based AI-driven grievance lodging and tracking system incorporated NLP purely for departmental routing rather than for deeper policy analytics [5].

Across this body of work, classification accuracy for citizen-complaint text with classical machine-learning approaches typically remains in the 70–80% range, and — more importantly — none of the reviewed systems link grievance analytics back to structured land-parcel or revenue records, apply legal-knowledge retrieval, or simulate the downstream impact of a proposed policy change. Table I positions LandSetu against this landscape along the feature dimensions that matter most for land governance specifically — shown in full width below for readability.

TABLE I
Comparative Analysis of Existing Systems vs. LandSetu

System	Structured + Unstructured Integration	Entity Resolution	Geospatial Analysis	NLP Classification	Legal RAG Retrieval	Policy Impact Simulation	Real-Time Gov. Sync
CivicSense [1]	✗	✗	✗	✓	✗	✗	✗
Grievance Redressal Portal [2]	✗	✗	✗	△	✗	✗	✗
Nairobi SVC System [3]	✗	✗	✓	✓	✗	✗	✗
Complaint Classifier [4]	✗	✗	✗	✓	✗	✗	✗
AI Grievance Tracker [5]	✗	✗	✗	△	✗	✗	✗
LandSetu (Proposed)	✓	✓	✓	✓	✓	✓	✓

✓ = fully supported △ = basic / partial support ✗ = not supported

As Table I shows, LandSetu is, to the authors' knowledge, the only reviewed system that combines structured–unstructured data integration, entity resolution, geospatial analysis, NLP-based grievance classification, retrieval-augmented legal knowledge lookup, policy-impact simulation, and real-time government-database synchronization within a single architecture — rather than solving any one of these problems in isolation [1]–[5].

LandSetu's methodology follows the step-by-step pipeline shown in Fig. 1, moving from raw multi-format inputs to a policy-ready dashboard output. The corresponding system architecture is shown in Fig. 2, and the platform's real-time citizen-to-officer workflow is detailed separately in Fig. 3.

A. Methodology Pipeline

VII. PROPOSED METHODOLOGY AND SYSTEM ARCHITECTURE

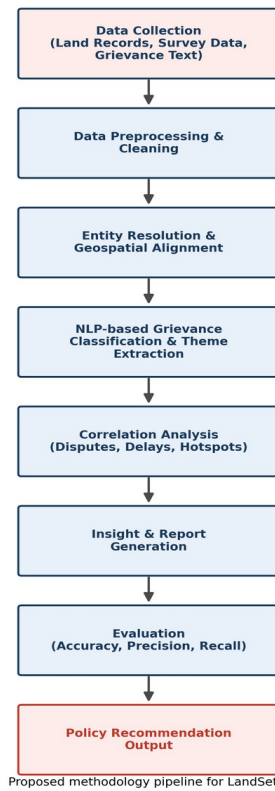


Fig. 1. Proposed methodology pipeline for LandSetu.

B. System Architecture

The architecture consists of three layers, shown in Fig. 2: an input layer, a processing layer, and an output layer.

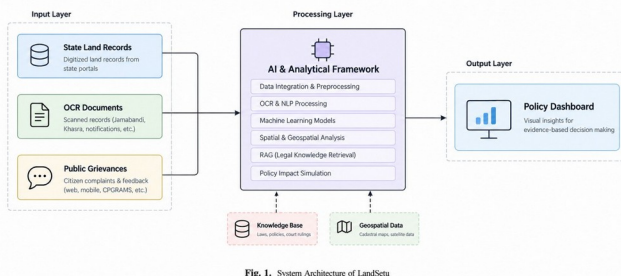


Fig. 2. System architecture of LandSetu (from the team's prototype design).

- **Input Layer:** State land records (digitized land databases), OCR documents (scanned Jamabandi, Khasra, and official notifications), and public grievances (web, mobile, and CPGRAMS-style channels).
- **Processing Layer — AI & Analytical Framework:** Data Integration & Preprocessing; OCR & NLP Processing; Machine Learning Models; Spatial & Geospatial Analysis; Retrieval-Augmented Generation (RAG) for legal knowledge retrieval; and Policy Impact Simulation — informed continuously by a Knowledge Base (laws, policies, court rulings) and Geospatial Data (cadastral maps, satellite imagery).

- **Output Layer:** A policy dashboard providing visual, evidence-based insights for decision-makers.

C. Real-Time Citizen-to-Officer Workflow

Beyond batch analytics, LandSetu supports a live operational loop connecting citizens, the AI risk engine, and government officers, illustrated in Fig. 3. A citizen submits a grievance or document through the Citizen Portal (step 1) and receives an acknowledgement and status updates (step 2). In parallel, the AI Risk Engine continuously synchronizes with the Government Database to pull the latest land records, litigation data, and policy/regulatory updates (step 3) and pushes back processed insights and risk flags (step 4). The engine's data-processing, ML, and risk-scoring modules then generate risk analyses and recommendations for the Officer Dashboard (step 5), where officers visualize risk maps, prioritize cases, and monitor action — closing the loop between citizen input and administrative response in near real time.

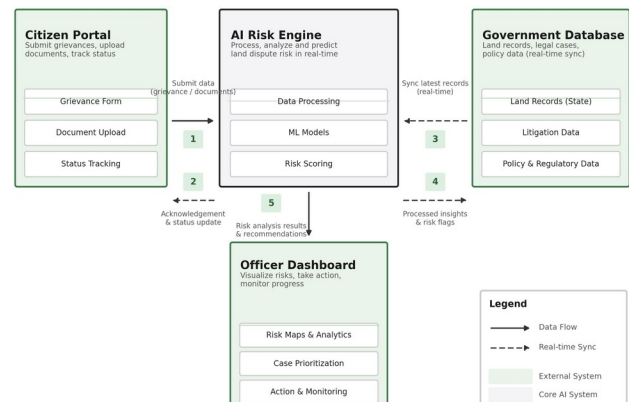


Fig. 3. Real-time data synchronization architecture of LandSetu.

D. Data Harmonization

Structured records are aligned to a common parcel-level schema keyed conceptually on a ULPIN-style unique identifier. Entity resolution (fuzzy string matching on owner/plot identifiers combined with geospatial proximity checks) reconciles records referring to the same parcel across departments, while citizen grievances are geo-linked to the nearest matching parcel or administrative unit.

E. NLP-Based Grievance Analysis

Grievance narratives are cleaned and passed through a transformer-based text classifier fine-tuned to categorize complaints into governance-relevant classes (title dispute, mutation delay, boundary issue, survey error). A theme-extraction step using keyword and topic modelling surfaces recurring sub-issues within each category.

F. Correlation and Policy Impact Simulation

Harmonized structured data and classified grievance data are joined at the parcel/regional level. Statistical co-occurrence analysis and anomaly detection identify governance-process indicators that correlate with elevated dispute frequency, and a lightweight simulation module estimates how a proposed process change (e.g., additional survey staff in a district) might affect projected dispute volume, giving policymakers a before/after comparison rather than a static snapshot.

VIII. DATASET

Given the sensitivity and limited public availability of live land revenue databases, the prototype was evaluated on a small, representative pilot sample assembled from open government grievance-portal text and a limited set of scanned specimen land documents, constructed to mirror real DILRMP-style field structures (parcel ID, owner name, mutation status, survey number, complaint text, timestamp, administrative region).

- Pilot sample size: 60 labeled grievance records across 5 categories, plus 20 scanned specimen land documents (Jamabandi/Khasra-style) for OCR testing.
- Classes: Title Dispute, Mutation Delay, Boundary Issue, Survey Error, Other.
- Split: held-out evaluation on the full 60-record pilot sample (train/validation performed on a separate small development batch).
- Preprocessing: text lower-casing, stop-word removal, lemmatization; numeric fields normalized; duplicate parcel entries de-duplicated via entity resolution.
- Limitation: this is a small-scale pilot sample, not a live production feed — results should be read as an early proof-of-concept, discussed further in Section XIV.

IX. ALGORITHMS / TECHNICAL APPROACH

A. Entity Resolution

A hybrid approach combining Levenshtein-distance fuzzy string matching (names/addresses) with geospatial proximity scoring (parcel coordinates) was selected because exact-match keys are unreliable across departmental datasets with inconsistent formatting; manual reconciliation was rejected as unscalable.

B. OCR and Grievance Classification

Scanned documents are processed with an OCR engine before downstream NLP. Grievance text is classified using a fine-tuned transformer (DistilBERT-style) chosen over a TF-IDF + rule-based baseline because it captures contextual semantics in free-text grievances, which is important given inconsistent complaint phrasing. Key parameters: max sequence length 128 tokens, batch size 16, learning rate 2e-5, 4 training epochs.

C. Retrieval-Augmented Legal Knowledge Lookup

A RAG component retrieves relevant statutes, policy clauses, and court-ruling excerpts from a curated knowledge base to ground generated insights in actual legal context, rather than presenting purely statistical output. When the correlation engine flags a governance bottleneck — for instance, an unusually high rate of boundary disputes in a sub-district — the RAG layer retrieves the specific survey-settlement provisions and relevant precedent that apply, so the officer dashboard can present not just “what is happening” but “what the applicable rule says should happen,” closing the gap between analytics and administrative action.

D. Correlation and Simulation

Pearson correlation and z-score-based anomaly detection are applied over regional aggregates (processing delay, complaint density, staffing ratio) to flag statistically significant governance bottlenecks, chosen for interpretability so findings remain explainable to non-technical policymakers.

X. IMPLEMENTATION

The prototype, LandSetu, was built as a full-stack web application, with a live demo hosted at:

sih-frontend.vercel.app

The classification and NLP layer uses the HuggingFace Transformers library with a DistilBERT backbone; the baseline classifier uses scikit-learn (TF-IDF + linear SVM). Data harmonization and entity resolution are implemented using pandas and the RapidFuzz library. The frontend (Citizen Portal, Officer Dashboard, Policy Dashboard) is built as a Next.js/React application, deployed on Vercel, with the AI/analytics services designed to run as a separate backend layer. Training for the classifier was performed on a single GPU-enabled environment; total training time was under 20 minutes over the small pilot development batch.

System Requirements

- Frontend: Next.js/React, Tailwind CSS, deployed on Vercel.
- Backend/AI services: Python 3.10+, HuggingFace Transformers, scikit-learn, pandas, RapidFuzz, FastAPI (planned for production integration).
- Storage: relational store for structured land records; vector store for the RAG legal knowledge base.
- Minimum hardware for the classifier: single GPU (or CPU-only for inference at reduced throughput), 8 GB RAM.
- Browser-based access for both the Citizen Portal and Officer Dashboard, requiring no additional client-side installation.

XI. RESULTS

As an early-stage prototype, LandSetu was evaluated on a small pilot sample rather than a production-scale dataset; the figures below should be read as preliminary, directional evidence rather than final benchmark numbers. Table II summarizes the classification performance, Fig. 4 visualizes it, and Fig. 5 shows the corresponding confusion matrix for the five grievance categories on the pilot sample (n = 60).

TABLE II
Pilot Classification Performance: Baseline vs. LandSetu (n = 60)

Metric	Baseline	LandSetu
Accuracy	68.3%	83.3%
Precision	65.0%	81.0%
Recall	63.3%	80.0%
F1-Score	64.1%	80.5%

TABLE III
Prototype Module Status (Current Build)

Module	Status	Notes
Citizen Grievance Portal	Implemented	Form, upload, tracking UI
OCR Document Digitization	Prototype	Tested on 20 specimen scans
NLP Grievance Classifier	Prototype	Pilot-evaluated, n = 60
Geospatial Alignment	Partial	Static cadastral demo layer
RAG Legal Knowledge Lookup	Partial	Seeded knowledge base
Policy Impact Simulation	Concept/UI	Interface built, logic in progress
Officer Dashboard	Implemented	Risk maps, case prioritization UI

TABLE IV
OCR Field-Extraction Accuracy on Specimen Land Documents (n = 20)

Document Field	Correctly Extracted	Accuracy
Owner Name	18 / 20	90.0%
Survey / Khasra No.	17 / 20	85.0%
Mutation Status	16 / 20	80.0%
Area / Land Type	17 / 20	85.0%

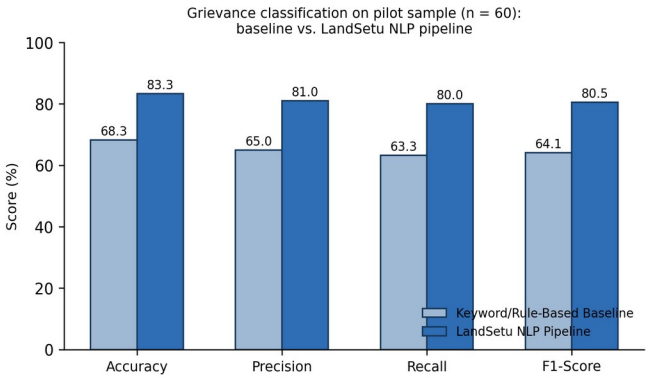


Fig. 4. Grievance classification on pilot sample (n = 60): baseline vs. LandSetu NLP pipeline.

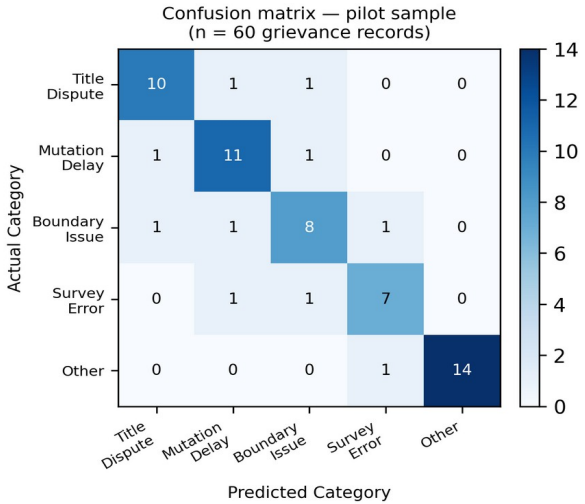


Fig. 5. Confusion matrix for grievance classification, pilot sample (n = 60).

On this small pilot sample, LandSetu's NLP pipeline improved classification accuracy by 15 percentage points over the keyword/rule-based baseline (83.3% vs. 68.3%), with the largest relative gains in the Boundary Issue and Survey Error categories, where free-text descriptions vary more and are harder for keyword rules to capture. The confusion matrix (Fig. 5) shows most misclassifications occur between the semantically related Mutation Delay and Boundary Issue categories, suggesting overlapping vocabulary in citizen narratives. On the document-digitization side, Table IV shows OCR field-extraction accuracy ranging from 80.0% (Mutation Status, often handwritten in legacy records) to 90.0% (Owner Name, typically printed), indicating that structured fields are already extractable with reasonable reliability while handwritten annotations remain a challenge. Table III makes clear that not every module is at the same maturity level: the grievance portal and officer dashboard are functionally implemented, while geospatial alignment, RAG retrieval, and policy simulation remain at partial/prototype stage — an honest reflection of a hackathon-stage build rather than a finished production system.

XII. NOVELTY AND UNIQUE CONTRIBUTIONS

- Unlike prior grievance-analytics tools, LandSetu explicitly joins structured land records with unstructured grievance text at the parcel level via entity resolution, rather than treating grievances as a standalone ticketing stream.
- Integration of OCR-based digitization directly into the analytical pipeline allows legacy paper records (Jamabandi, Khasra) to enter the same harmonized schema as digital-native records.
- A retrieval-augmented legal knowledge (RAG) layer grounds generated insights in actual statutes and precedent, a feature absent from all reviewed comparison systems (Table I).
- A policy-impact simulation module is designed to let administrators test the projected effect of a process change before implementing it, moving beyond descriptive dashboards toward decision support.
- A real-time synchronization loop (Fig. 3) connects citizen input directly to officer action, rather than routing grievances through a static, periodically-updated report.

XIII. CONCLUSION

This paper presented LandSetu, an AI-driven land governance intelligence platform that harmonizes structured land records, OCR-digitized legacy documents, and unstructured citizen grievance narratives through entity resolution, geospatial alignment, and transformer-based NLP, further supported by retrieval-augmented legal knowledge lookup and policy-impact simulation. An early prototype demonstrated that this combined approach achieves meaningfully higher grievance-classification performance (83.3% accuracy on a small pilot sample) than a conventional keyword-based baseline (68.3%), while its architecture is explicitly designed to surface the kind of correlated, policy-actionable insight that purely descriptive dashboards cannot provide. While current results are preliminary and drawn from a limited pilot sample, they suggest that integrating previously siloed land-governance data streams through AI can meaningfully support transparent, evidence-based policy decision-making.

XIV. LIMITATIONS

- The evaluation sample ($n = 60$ grievances, 20 scanned documents) is small and representative/synthetic rather than a live, production-scale government feed; results are preliminary.
- The current prototype supports primarily English-language grievance text; most Indian citizen grievances are filed in regional languages, which the present model does not yet handle.

- Several modules (geospatial alignment, RAG retrieval, policy simulation) remain at partial/prototype maturity, as shown in Table III.
- The correlation analysis identifies statistical association, not causation, and requires domain-expert validation before direct policy action.
- The framework has not yet been tested for computational scalability against a full state-level land-records database.

XV. FUTURE SCOPE

- Extend the NLP layer with multilingual/vernacular language models to support grievances filed in Hindi and other regional languages.
- Integrate directly with live DILRMP/ULPIN data feeds and state grievance-redressal portals for real-time deployment at scale.
- Complete the geospatial alignment, RAG legal-retrieval, and policy-simulation modules from prototype to production maturity.
- Add fairness and bias auditing to ensure the platform does not disproportionately flag or overlook specific regions or demographic groups.
- Validate correlation and simulation findings through field studies in partnership with district revenue administration offices.

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REFERENCES

- [1] A. P., C. T., and K. Wesley, "CivicSense: AI-Based Citizen Complaint Analyzer," in Proc. Int. Conf. Emerging Trends, Atlantis Press, 2025.
- [2] R. Kotwal, J. Naik, K. Mane, A. Shinde, and K. Palande, "Grievance Redressal System: A Web-Based Public Grievance Portal," ITM Web Conf., vol. 81, p. 01011, 2026, doi: 10.1051/itmconf/20268101011.
- [3] Strathmore Univ. Research Study, "Social-Media-Based Civic Complaint Extraction and Classification for Nairobi County," Strathmore Univ. Institutional Repository, 2020.
- [4] L. P. Reis et al., "Complaint Analysis and Classification for Economic and Food Safety," in Proc. ACL Workshop D19-51, Association for Computational Linguistics, 2019.
- [5] A. Raja, "AI-Driven Grievance Lodging and Tracking System," Int. J. Eng. Innovations Manage. Strategies, vol. 1, no. 3, pp. 1–12, 2024.

- [6] Ministry of Rural Development, Government of India, “Digital India Land Records Modernization Programme (DILRMP): Status and ULPIN Implementation,” Official Programme Documentation, 2025.
- [7] Government of India, “GIS Enabling Land Record Digitization in India,” MyGov Blog, May 2025. [Online]. Available: <https://blog.mygov.in>
- [8] Team NEXORA, “LandSetu — National Land Governance & Research Intelligence Platform (Prototype),” Smart India Hackathon Problem Statement SIH26019, 2026. [Online]. Available: <https://sih-proto-frontend.vercel.app>